

Review of Integration + Tables

Eg $\int \cos(2x) dx = \frac{1}{2} \int \cos u du = \frac{1}{2} \sin(2x) + C$

Eg $\int \frac{t}{1+(2t)^4} dt = \int \frac{t}{1+(2t)^2} dt = \frac{1}{2} \tan^{-1}(2t) + C$

Eg $\int \frac{6u^2}{4+u} du$ There is no obvious substitution, try tables

SBS Appendix D #35 $\int \frac{u^2}{au+b} du = \frac{u}{a} - \frac{b}{a^2} \ln|au+b|$

$6(u - b \ln|u+4|) + C$

Important The FTC implies all continuous functions have an antiderivative, but even simple looking ones may not have a closed form.

In practice, numerical methods are very important.

Integration by Parts

$$\int u dv = uv - \int v du$$

This is the integral version of the product rule: ~~(uv)'~~ $(uv)' = u dv + v du$

Eg $\int x \sin x dx$

$$\begin{array}{lcl} u = x & \searrow & dv = \sin x dx \\ du = dx & \longrightarrow & v = -\cos x \end{array}$$

$$-x \cos x - \int -\cos x dx = -x \cos x + \sin x + C$$

Idea u = part that gets better when we take derivative
 v = part that we can integrate

Examples
 $x, x^n, \ln x$
 $\sin x, \cos x, e^x$

Eg $\int x^2 e^{5x} dx$

$$\begin{array}{lcl} u = x^2 & & dv = e^{5x} dx \\ du = 2x dx & & v = \frac{1}{5} e^{5x} \end{array}$$

$$\frac{1}{5} x^2 e^{5x} - \int \frac{2}{5} x e^{5x} dx$$

$$\begin{array}{l} u = x \\ du = dx \end{array}$$

$$\begin{array}{l} dv = e^{5x} dx \\ v = \frac{1}{5} e^{5x} \end{array}$$

$$\frac{1}{5} x^2 e^{5x} - \frac{2}{5} \left(\frac{1}{5} x e^{5x} - \frac{1}{25} e^{5x} \right) + C$$

Eg $\int x^3 \sin(x) dx$

There is a simpler way to keep track:

<u>Derivatives</u>		<u>Integrals</u>
x^3	+	$\sin x$
$3x^2$	-	$-\cos x$
$6x$	+	$-\sin x$
6	-	$\cos x$
0		$\sin x$

$$x^3(-\cos x) - 3x^2(-\sin x) + 6x \cos x - 6 \sin x + C$$

Eg $\int e^{3x} \cos x dx$

Let $u = e^{3x}$

$dv = \cos x dx$

$du = 3e^{3x} dx$

$v = \sin x$

$$e^{3x} \sin x - \int 3e^{3x} \sin x dx$$

So $\boxed{\int e^{3x} \cos x dx} = e^{3x} \sin x - 9e^{3x}(\cos x) + 9 \boxed{\int e^{3x} \cos x dx}$

$$\int e^{3x} \cos x dx = \frac{1}{10} (e^{3x} \sin x + 9e^{3x}) + C$$

Eg $\int \ln x \, dx$

Not much choice....

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$dv = 1 \, dx$$

$$v = x$$

$$x \ln x - \int dx = x \ln x - x + C$$

Eg $\int_0^1 \tan^{-1} x \, dx$

$$u = \tan^{-1} x$$

$$du = \frac{1}{1+x^2} dx$$

$$dv = dx$$

$$v = x$$

$$x \tan^{-1} x - \int \frac{x}{1+x^2} dx$$

$$\left(x \tan^{-1} x - \frac{1}{2} \ln |1+x^2| \right) \Big|_0^1 = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

Powers of Trig Functions

Key Identities:

$$\cos^2 x + \sin^2 x = 1$$

$$\cos^2 x = \frac{1}{2} (1 + \cos 2x)$$

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

We can turn $\sin^2 x$ into $\cos^2 x$ and vice versa and integrate both

Eg $\int \cos^2 x \sin^3 x \, dx = \int \cos^2 x (1 - \cos^2 x) \sin x \, dx$

$$-\int u^2 (1 - u^2) \, du = -\frac{u^3}{3} + \frac{u^5}{5} + C = -\frac{1}{3} \cos^3 x + \frac{1}{5} \cos^5 x + C$$

Sec x, tan x

We can do some with $1 + \tan^2 x = \sec^2 x$ and

$$\int \tan x \, dx = \ln |\sec x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

Try Substitution

Start with

$$\sqrt{a^2 - x^2}$$

$$\sqrt{x^2 + a^2}$$

$$\sqrt{x^2 - a^2}$$

Let

$$x = a \sin \theta$$

$$x = a \tan \theta$$

$$x = a \sec \theta$$

Get

$$a \cos \theta$$

$$a \sec \theta$$

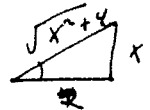
$$a \tan \theta$$

Eg $\int \frac{dx}{\sqrt{4+x^2}}$

Let $x = 2 \tan \theta$. $dx = 2 \sec^2 \theta d\theta$

$$\int \frac{1}{2 \sec \theta} 2 \sec^2 \theta d\theta = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C$$

$$\ln \left| \sec \left(\tan^{-1} \frac{x}{2} \right) + \tan \left(\tan^{-1} \frac{x}{2} \right) \right| + C$$

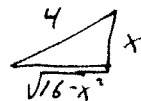


$$\ln \left| \frac{\sqrt{x^2 + 4}}{2} + \frac{x}{2} \right| + C$$

Eg $\int \frac{x}{\sqrt{16-x^2}} dx$

Let $x = 4 \sin \theta$. $dx = 4 \cos \theta d\theta$

$$\int \frac{4 \sin \theta}{4 \cos \theta} (4 \cos \theta) d\theta = \int 4 \sin \theta d\theta = -4 \cos \left(\sin^{-1} \frac{x}{4} \right) + C$$

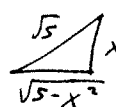


$$= -4 \left(\frac{\sqrt{16-x^2}}{4} \right) + C$$

Eg $\int \frac{x^2}{\sqrt{5-x^2}} dx$ Let $x = \sqrt{5} \sin \theta$ $dx = \sqrt{5} \cos \theta d\theta$

$$\int \frac{5 \sin^2 \theta}{\sqrt{5} \cos \theta} \sqrt{5} \cos \theta d\theta = \int 5 \sin^2 \theta d\theta = \int \frac{5}{2} (1 - \cos 2\theta) d\theta$$

$$\frac{5}{2} \theta - \frac{5}{4} \sin(2\theta) + C = \frac{5}{2} \sin^{-1}\left(\frac{x}{\sqrt{5}}\right) - \frac{5}{4} \sin\left(2 \sin^{-1} \frac{x}{\sqrt{5}}\right)$$

$$\frac{5}{2} \sin^{-1}\left(\frac{x}{\sqrt{5}}\right) - \frac{5}{2} \sin\left(\sin^{-1} \frac{x}{\sqrt{5}}\right) \cos\left(\sin^{-1} \frac{x}{\sqrt{5}}\right) + C$$


$$\frac{5}{2} \sin^{-1}\left(\frac{x}{\sqrt{5}}\right) - \frac{5}{2} \left(\frac{x}{\sqrt{5}}\right) \left(\frac{\sqrt{5-x^2}}{\sqrt{5}}\right) + C$$

Aside $\sqrt{x^2+3x-5} = \sqrt{\left(x^2+3x+\frac{9}{4}\right) - \frac{29}{4}} = \sqrt{\left(x+\frac{3}{2}\right)^2 - \frac{29}{4}}$

So if $\int$ were in an integral, we'd let $u = x + \frac{3}{2}$ and go from there.

Important Formula $\int \frac{1}{x^2+a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$

Let $x = a \tan \theta$ $dx = a \sec^2 \theta d\theta$

$$\int \frac{1}{a^2 \sec^2 \theta} a \sec^2 \theta d\theta = \frac{1}{a} \theta + C = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right).$$

Partial Fractions

With a couple of algebra tricks, we can integrate any rational functions.

① Long Division - This works if $\deg \text{top} \geq \deg \text{bottom}$

Eg $\int \frac{x^3 + x}{x-1} dx$

$$\begin{array}{r} x^2 + x + 2 \\ x-1 \overline{) x^3 + + x} \\ \underline{x^3 - x^2} \\ x^2 + x \\ \underline{x^2 - x} \\ 2x \\ \underline{2x - 2} \\ 2 \end{array}$$

$$\int x^2 + x + 2 + \frac{2}{x-1} dx$$

$$\frac{x^3}{3} + \frac{x^2}{2} + 2x + 2 \ln |x-1| + C$$

② Partial fractions - Turn division into addition

Eg $\int \frac{3x-2}{(x+2)(x-1)} dx = \int \frac{A}{x+2} + \frac{B}{x-1} dx$

$$\frac{A(x-1) + B(x+2)}{(x+2)(x-1)} = \frac{3x-2}{(x+2)(x-1)} \Rightarrow A(x-1) + B(x+2) = 3x-2$$

① $(A+B)x - A + 2B = 3x - 2$

$$\begin{aligned} A+B &= 3 \\ -A+2B &= -2 \end{aligned}$$

$$3B = 1 \Rightarrow B = \frac{1}{3} \quad A = \frac{8}{3}$$

② Heaviside Method $A(x-1) + B(x+2) = 3x-2$

This equation is true $\forall x \in \mathbb{R}$, so try a few specific x 's to "strike" factors

$$\begin{array}{lll} \underline{x=1} & 3B = 1 & \Rightarrow B = \frac{1}{3} \\ \underline{x=-2} & -3A = -8 & \Rightarrow A = \frac{8}{3} \end{array}$$

$$\int \frac{8/3}{x+2} + \frac{1/3}{x-1} dx = \frac{8}{3} \ln|x+2| + \frac{1}{3} \ln|x-1| + C$$

4+
set

Eg $\int \frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3} dx = \int 2x + \frac{5x-3}{(x+1)(x-3)} dx$

$$\frac{A}{x+1} + \frac{B}{x-3} \Rightarrow A = 2 \quad B = 3$$

$$x^2 + 2 \ln|x+1| + 3 \ln|x-3| + C$$

$$\text{Eg } \int \frac{2x^2 - x + 4}{x^3 + 4x} dx = \int \frac{2x^2 - x + 4}{x(x^2 + 4)} dx$$

$x^2 + 4$ is an irreducible quadratic

$$\frac{A}{x} + \frac{Bx+C}{x^2+4} \Rightarrow A(x^2+4) + Bx^2 + Cx = 2x^2 - x + 4$$

$$(A+B) = 2$$

$$C = -1$$

$$A = 1$$

$$B = 1$$

$$C = -1$$

$$4A = 4$$

$$\int \frac{1}{x} + \frac{x}{x^2+4} + \frac{-1}{x^2+4} dx$$

$$\ln |x| + \frac{1}{2} \ln |x^2+4| - \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C$$

Note Heaviside isn't as much use here

$x=0$ will get rid of B and C , but nothing will get rid of x^2+4

- General Rules
- ① Linear terms get constants A
 - ② Quadratic terms get linear $Bx+C$
 - ③ Multiple terms get repeated

$$\text{Eg } \frac{x^5 - 7x^3 + 5}{(x-2)(x^2+x+7)(x+4)^3(x^2+1)^2}$$

$$\frac{A}{x-2} + \frac{Bx+C}{x^2+x+7} + \frac{D}{x+4} + \frac{E}{(x+4)^2} + \frac{F}{(x+4)^3} + \frac{Gx+H}{x^2+1} + \frac{Ix+J}{(x^2+1)^2}$$

$$\underline{\text{Eg}} \quad \int \frac{4x}{x^3 - x + 1} dx = \int \frac{4x}{(x+1)(x-1)^2} dx = \int \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2} dx$$

$$A(x-1)^2 + B(x+1)(x-1) + C(x+1) = 4x$$

$$\underline{x=1} \quad 2C = 4 \quad \Rightarrow C = 2$$

$$\underline{x=-1} \quad 4A = -4 \quad \Rightarrow A = -1$$

$$\underline{x=0} \quad A - B + C = 0 \quad \Rightarrow B = 1$$

$$\int \frac{-1}{x+1} + \frac{1}{x-1} + \frac{2}{(x-1)^2} dx$$

$$-\ln|x+1| + \ln|x-1| + 2 \frac{(x-1)^{-1}}{-1} + C$$

$$\underline{\text{Eg}} \quad \int \frac{x-1}{x^2+2x+3} dx \quad \text{Irreducible, Complete Square} \quad \int \frac{x-1}{(x+1)^2+2} dx$$

$$\text{Let } u = x+1, \quad \int \frac{x-1}{u^2+2} du = \int \frac{(u-1)-1}{u^2+2} du$$

$$\int \frac{u}{u^2+2} du + \int \frac{-2}{u^2+2} du$$

$$\frac{1}{2} \ln|u^2+2| - \frac{2}{\sqrt{2}} \tan^{-1}\left(\frac{u}{\sqrt{2}}\right) + C$$

$$\frac{1}{2} \ln|(x+1)^2+2| - \frac{2}{\sqrt{2}} \tan^{-1}\left(\frac{x+1}{\sqrt{2}}\right) + C$$

Summary

There is a summary table in text p. 458 SBS.

1st Order Linear ODE's

Defn A differential eq. is eq. w/ derivatives in it.

Eg $\frac{dy}{dx} + \frac{3}{x}y = \cos(x^4) + 5$ is a 1st order linear ODE

The secret to solving is using the integrating factor $e^{\int \frac{3}{x} dx}$ (no c)

$$e^{\int \frac{3}{x} dx} = e^{3 \ln|x|} = e^{\ln|x|^3} = x^3$$

$$\underbrace{x^3 \frac{dy}{dx} + x^3 \frac{3}{x}y}_{\substack{\uparrow \\ \text{clearly the result of prod. rule} \\ \text{derivative of } x^3 y.}} = x^3 \cos(x^4) + 5x^3$$

$$x^3 y = \int x^3 \cos(x^4) + 5x^3 dx = \frac{1}{4} \sin(x^4) + \frac{5}{4} x^4 + C$$

$$y = \frac{1}{x^3} \left(\frac{1}{4} \sin(x^4) + \frac{5}{4} x^4 + C \right)$$

Eg Solve $\frac{1}{2x} \frac{dy}{dx} + y = \frac{1}{2x} e^{3x-x^2}$

Rewrite: $\frac{dy}{dx} + 2x y = e^{3x-x^2}$

Int factor: $e^{\int 2x dx} = e^{x^2}$

$$e^{x^2} \frac{dy}{dx} + e^{x^2} 2x y = e^{3x}$$

$$e^{x^2} y = \frac{1}{3} e^{3x} + C \Rightarrow y = e^{-x^2} \left(\frac{1}{3} e^{3x} + C \right)$$

IVP Find solution if $y(0) = 4$.

$$4 = y(0) = \frac{1}{3} + C \Rightarrow C = \frac{11}{3}$$

$$y = e^{-x^2} \left(\frac{1}{3} e^{3x} + \frac{11}{3} \right)$$

Hyperbolic Trig

Defn $\cosh x = \frac{e^x + e^{-x}}{2}$ $\sinh x = \frac{e^x - e^{-x}}{2}$

Defn $\operatorname{sech} x = \frac{1}{\cosh x}$ $\tanh x = \frac{\sinh x}{\cosh x}$ etc.

Why call them after trig functions?

$\frac{d}{dx} \cosh x = \sinh x$

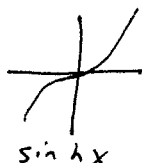
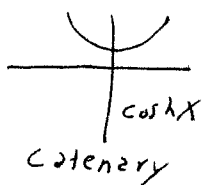
$\frac{d}{dx} \sinh x = \cosh x$

$\frac{d}{dx} \tanh x = \operatorname{sech}^2 x$

Ex $\int 3x \sinh(4x^2) = \frac{3}{8} \cosh(4x^2) + C$

Identities $\cosh^2 x - \sinh^2 x = 1$

$\sinh(2x) = 2 \sinh x \cosh x$



$\cosh(0) = 1$

$\sinh(0) = 0$

Real Reason for Similarity $\cos x = \frac{e^{ix} + e^{-ix}}{2}$

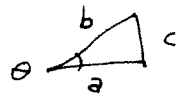
$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$

ODE $y'' = -y$ vs. $y'' = y$

Geometry Euclidean $c^2 = a^2 + b^2 - 2ab \cos \theta$

Spherical $\cos c = \cos a \cos b - \sin a \sin b \cos \theta$

Hyperbolic $\cosh c = \cosh a \cosh b - \sinh a \sinh b \cos \theta$



Engineering A $\cosh x$ graph is a catenary. Electric power lines hanging and the St. Louis Arch are catenaries.

Improper Integrals

If $\int_1^3 \frac{1}{x^2} dx$ makes sense as an area, what about $\int_1^{\infty} \frac{1}{x^2} dx$?

Eg / Defn $\int_1^{\infty} \frac{1}{x^2} dx = \lim_{N \rightarrow \infty} \left(\int_1^N \frac{1}{x^2} dx \right) = \lim_{N \rightarrow \infty} \left(-\frac{1}{N} - \frac{-1}{1} \right) = 1$

Eg $\int_1^{\infty} \frac{1}{x} dx = \lim_{N \rightarrow \infty} \ln N - \ln(1) = \infty$

If the limit is finite, we say the improper integral converges.
otherwise, the limit diverges.

Notice that if the integral is going to converge 0 must be a horizontal asymptote and must get small quickly

Prop $\int_1^{\infty} \frac{1}{x^p} dx$ converges $\Leftrightarrow p > 1$

Eg $\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{N \rightarrow 0} 2\sqrt{1} - 2\sqrt{N} = 2$ converges

Vertical Asymptotes provide another kind of improper integral.
You always have to check to be sure integrand is continuous.

Eg $\int_0^3 \frac{dx}{x-1}$



$$\lim_{N \rightarrow 1^+} \int_0^N \frac{dx}{x-1} + \int_N^3 \frac{dx}{x-1}$$

$$\lim_{N \rightarrow 1^+} \ln|N-1| - \ln|N-1| + \ln|2| \quad \underline{\underline{DNE}}$$

Divergent, but if we ignored the asymptote, we'd get
 $\ln|x-1| \Big|_0^3 = \ln 2$

Note $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$ makes sense as $\int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx$

Gabriel's Horn Rotate $f(x) = \frac{1}{x}$, $x \geq 1$ about x -axis 

$$\text{Volume} = \int_1^{\infty} \pi \left(\frac{1}{x}\right)^2 dx = \pi$$

$$\text{Surface Area} = \int_1^{\infty} 2\pi \left(\frac{1}{x}\right) \sqrt{1 + \left(\frac{-1}{x^2}\right)^2} dx \quad \underline{\underline{\text{Diverges!}}}$$

This is a paint can that can be filled with π unit³ of paint,
but the surface requires an infinite amount of paint.